AI Clinical Triage System

Project Overview Document

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1. Executive Summary

Our AI Clinical Triage System is a **voice-first, multilingual clinical triage platform** designed to automate patient intake at hospitals and clinics. The system conducts natural voice conversations with patients in their preferred language, collects symptoms, scores them against a disease cluster database using an intelligent triage scoring engine, and routes patients to the appropriate specialist — all in real-time, without human intervention.

The platform currently supports **6 languages** (Urdu, English, Hindi, Arabic, Punjabi, Malay), **12 medical specialties**, **70+ medical conditions**, **199 symptoms with 173 Urdu/English aliases**, and delivers end-to-end voice response times under **5 seconds**.

Key Metrics

Metric	Value
Languages Supported	6 (expandable to 30+)
Medical Specialties	12 departments
Medical Conditions	70+ (disease clusters with ICD-10 codes)
Symptom Aliases	173 entries covering 199 condition symptoms
Triage Scoring Tests	188 tests, 100% pass rate
Voice Response Time	3-5 seconds end-to-end
STT Options	Deepgram Nova-3, OpenAI gpt-4o-transcribe
STT Noise Filtering	Two-layer: Silero VAD (0.80) + Deepgram confidence (0.70)
Concurrent Sessions	Scalable via cloud deployment

Metric	Value
Availability	24/7/365
Total Automated Tests	265 tests, 100% pass rate

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2. Problem Statement

The Hospital Reception Bottleneck

Hospitals in South Asia, the Middle East, and Southeast Asia face critical challenges at their front desk:

1. **Long wait times** — Patients wait 15-45 minutes for basic registration and triage, especially during peak hours.
2. **Language barriers** — Multilingual populations (Pakistan: Urdu/Punjabi/English; Malaysia: Malay/English/Mandarin; UAE: Arabic/Urdu/Hindi/English) create communication gaps with reception staff.
3. **Inconsistent triage** — Junior receptionists lack medical knowledge to properly route patients, leading to wrong department visits and wasted time.
4. **Staff shortages** — Qualified medical receptionists are expensive and hard to retain, especially for 24/7 coverage.
5. **No documentation** — Verbal interactions at reception are lost — no record of initial complaints for the doctor.
6. **Peak hour collapse** — Emergency situations overwhelm reception, delaying care for non-emergency patients.

Market Size

Market	2024 Value	Projected	CAGR
Global Healthcare AI	$14.9B	$110.6B by 2030	38.6%
Conversational AI in Healthcare	$13.7B	$48.9B by 2030	23.8%
Middle East Healthcare AI	$436M	$8.4B by 2033	37.0%
Middle East Hospital Services	$114B	$172.8B by 2033	4.7%

Sources: Grand View Research, MarketsandMarkets, BCC Research (2024-2025)

Target Market Infrastructure:

Country	Hospitals	Healthcare Market	Key Driver
Pakistan	1,830 (1,147 public + 683 private)	$7B	240M population, 4+ languages
Saudi Arabia	2,590 (290 hospitals + 2,300 centers)	$67B	Vision 2030 — $65B investment
UAE	972	$22B → $50B by 2029	8% CAGR, premium spend
Qatar	100+	$12B	#1 Arab world healthcare ranking
Malaysia	355 (148 public + 207 private)	$28B → $48B by 2028	MyDigital, 13.5% CAGR

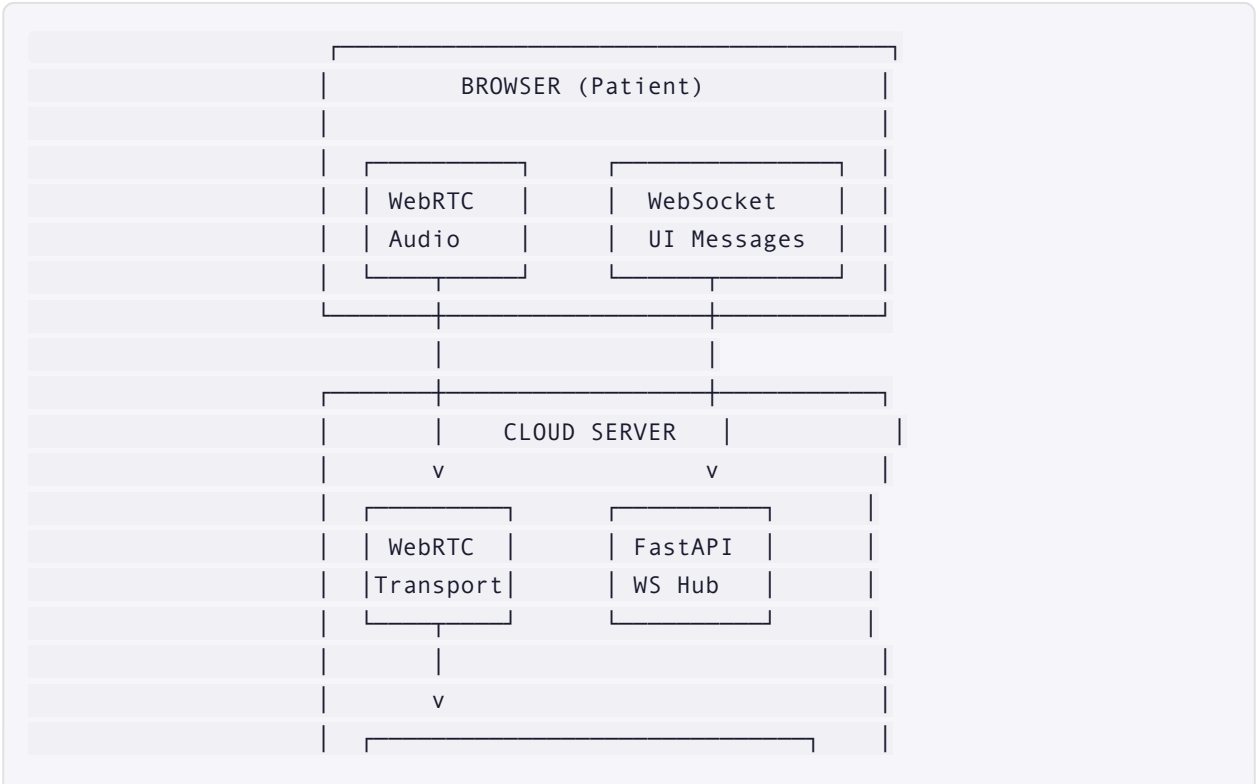
- **<5% of hospitals** in these regions have any form of AI automation at reception
- **Average hospital spends $180K-$350K/year** on front desk staffing (3-shift coverage)

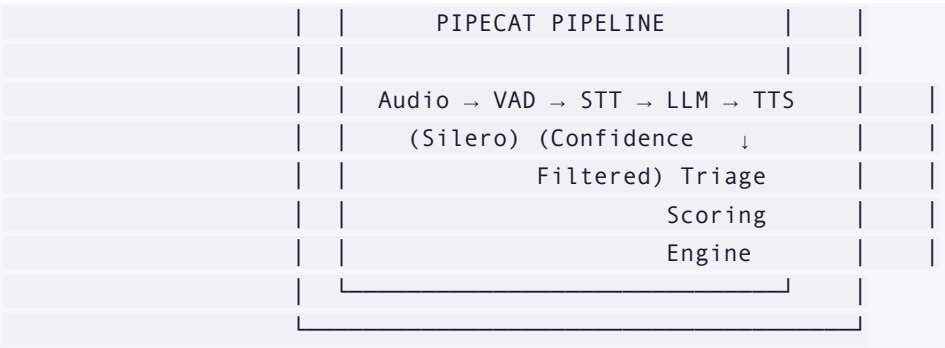
Sources: WHO EMRO, Statista, US-Saudi Business Council, UAE Open Data, Malaysia MOH (2024)

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3. Solution Architecture

3.1 System Overview





3.2 Technology Stack

Layer	Technology	Why
Voice Activity Detection	Silero VAD (neural ONNX)	Filters background noise — confidence=0.6, min_volume=0.5, stop_secs=0.6
Speech-to-Text	ConfidenceFilteredDeepgramSTT / OpenAI gpt-4o-transcribe	Dual STT with confidence-based noise filtering (drops < 0.70). Switchable from settings
Language Model	OpenAI GPT-4o-mini / GPT-4o / Claude	Follows triage directive — MUST refer to highest-scoring department
Triage Scoring Engine	Custom Python (disease_clusters.json)	70+ conditions, 173 aliases, exact matching, cross-condition boost, age/gender rules
Text-to-Speech	OpenAI TTS-1 / ElevenLabs Flash v2.5	Multi-sentence audio caching, number-to-Urdu conversion for Urdu TTS
Audio Transport	WebRTC (STUN/TURN)	Real-time bidirectional audio with built-in echo cancellation. Works through NAT/firewalls
Pipeline	Pipecat-AI 0.0.85	Async frame-based pipeline: VAD → STT → Broadcaster → LLM → Response → TTS Cache → TTS → Audio
Backend	FastAPI (Python)	HTTP, WebSocket, WebRTC endpoints. Tool-based LLM integration with patient screening
Frontend	Single-file HTML/CSS/JS	Real-time triage dashboard with Urdu tooltips, patient screening card, cost analysis

3.3 Triage Scoring Engine

The core intelligence of the system — a real-time scoring algorithm that matches patient symptoms against 70+ medical conditions organized in 12 departments.

Scoring Pipeline:

1. **Symptom Alias Expansion** — Patient symptoms (in any language/script) are expanded via 173-entry alias lookup. "کھانسی" → "cough", "سینے میں درد" → "chest pain", "نیند نہیں آتی" → "insomnia"
2. **Per-Original-Symptom Matching** — Each patient symptom matches at most ONE condition symptom, preventing alias chain inflation
3. **Exact Word-Set Matching** — "cough" matches "cough" but NOT "persistent cough" (clinically different symptoms)
4. **Vital Boost** — +10% per matching vital (BP, heart rate, temperature, sugar, mobility, fever)
5. **Severity Boost** — +5% when patient severity matches condition severity levels
6. **Cross-Condition Boost** — When symptoms match multiple conditions in a department, confidence increases: `best_score + best_score × 0.15 × (strong_conditions - 1)`
7. **Department Selection Rules** — Age-gated: Pediatrics zeroed for adults (≥12). Gender-gated: Gynecology zeroed for males. Unknown age: 70% penalty on Pediatrics unless child-specific symptoms present
8. **Doctor Availability Check** — Unavailable doctors skipped in referral directive; [NOT AVAILABLE] tag shown
9. **Directive Referral** — Triage returns explicit "REFER TO → Dr. Name, Department, Room" — the LLM MUST follow this directive

Scoring Formula:

```
condition_score = (matched / total_symptoms) + vital_boost + severity_boost
department_score = best_condition_score + cross_condition_boost
final_score = apply_department_rules(department_score, age, gender)
```

Noise Filtering (Three-Layer Defense, no audio processing filters):

Layer	Technology	Threshold	Purpose
Audio Level	Silero VAD (neural)	confidence=0.6, min_volume=0.5, stop_secs=0.6	Filters non-speech audio
STT Level	ConfidenceFilteredDeepgramSTT	confidence < 0.45 dropped	

Layer	Technology	Threshold	Purpose
			Filters low-confidence transcriptions (Urdu-friendly)
Text Level	Noise word filter	frozenset lookup	Drops "hmm", "ok", "yeah" etc.

Pipeline Performance:
- All UI broadcasts: non-blocking (fire-and-forget)
- File I/O (_persist): thread pool
- Scoring: skipped on greeting/name turns
- allow_interruptions=true (STT always running)
- Pipeline overhead: < 1ms per turn

3.4 Conversation Flow

```
1.  Patient opens browser → WebRTC audio connected
2.  Agent greets patient in selected language (pure Urdu script, no English mixing)
3.  Patient speaks → Silero VAD filters background noise → Confidence-filtered STT
4.  LLM processes speech → Calls update_patient_screening tool on EVERY response
5.  Triage scoring engine scores symptoms against 70+ conditions in real-time
6.  Triage dashboard updates live — department confidence percentages shown
7.  Agent collects: Name → Symptoms → Duration → Vitals (symptom-relevant only)
8.  Smart follow-up: asks targeted symptoms to differentiate conditions (e.g., chest pain → "ہے؟
9.  Off-topic answer handling: if patient answers a different question, agent re-asks
10. TTS converts response → Number-to-Urdu conversion → Audio caching → WebRTC output
11. When triage score ≥ 50%: agent follows "REFER TO →" directive from scoring engine
12. Agent provides: Doctor name, department, room number (availability checked)
13. Patient screening card + triage slip available for printing
```

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4. Core Features

4.1 Multilingual Voice Interaction

The system natively handles 6 languages with automatic text direction (RTL/LTR) support:

Language	Region	Direction	Script	Cultural Adaptation
Urdu	Pakistan, India	RTL	Nastaliq	آپ/جناب forms, Islamic greetings
English	Global	LTR	Latin	Professional medical terminology

Language	Region	Direction	Script	Cultural Adaptation
Hindi	India	LTR	Devanagari	Respectful "aap" forms
Arabic	Middle East	RTL	Arabic	Islamic honorifics, Gulf dialect awareness
Punjabi	Pakistan, India	RTL	Shahmukhi	Regional expressions, cultural warmth
Malay	Malaysia, Indonesia	LTR	Latin	Formal "Encik/Puan" honorifics

4.2 Configurable Agent Personas

10 pre-built personas with gender-appropriate voice, grammar, and cultural behavior:

Persona	Gender	Voice Provider	Voice	Use Case
Amina	Female	OpenAI	Nova	Default — warm, caring receptionist
Ayesha	Female	OpenAI	Shimmer	Professional, efficient
Fatima	Female	OpenAI	Alloy	Friendly, approachable
Hira	Female	ElevenLabs	Rachel	Premium natural voice
Sarah	Female	ElevenLabs	Sarah	Gentle, reassuring
Ali	Male	OpenAI	Echo	Professional male agent
Hassan	Male	OpenAI	Onyx	Deep, authoritative voice
Ahmed	Male	OpenAI	Fable	Warm, storyteller quality
Bilal	Male	ElevenLabs	Drew	Natural conversational
Usman	Male	ElevenLabs	Paul	Calm, measured delivery

Each persona adapts:
- **Grammatical gender** — Urdu masculine/feminine verb endings
- **Cultural honorifics** — Gender-appropriate terms of address
- **Voice characteristics** — Matching TTS voice and tone
- **Personality traits** — Warm, Professional, or Friendly

4.3 Real-Time Disease Cluster Triage

The triage scoring engine scores patient symptoms in real-time against a disease cluster database. Each department shows a live confidence percentage on the triage dashboard. The

system maps patient symptoms (including Urdu aliases) to 12 specialist departments and 70+ conditions:

Symptom Category	Specialist	Room
Fever, flu, general illness	General Medicine	101
Chest pain, blood pressure	Cardiology	302
Pregnancy, women's health	Gynecology	205
Children's illness, vaccination	Pediatrics	108
Bone fractures, joint pain	Orthopedics	210
Skin rash, allergies	Dermatology	115
Ear, nose, throat	ENT	118
Eye problems, vision	Ophthalmology	122
Headaches, seizures	Neurology	310
Toothache, dental	Dentistry	G-05
Stomach pain, digestion	Gastroenterology	215
Anxiety, depression	Psychiatry	320
Severe/Life-threatening	**Emergency**	**Ground Floor**

The agent checks doctor availability in real-time and suggests alternatives when a specialist is unavailable.

Triage Dashboard Features:
- **Live confidence percentages** per department — updates with each new symptom
- **Expandable condition cards** — click to see individual condition scores, matched/unmatched symptom pills, vital requirements, severity levels
- **Urdu tooltips** — hover over any English symptom tag to see Urdu translation (all 199 symptoms covered)
- **Matched symptom highlighting** — green pills for matched symptoms, gray for unmatched
- **ICD-10 codes** — each condition shows its ICD-10 classification code
- **Cross-condition boost indicator** — shows when multiple conditions in a department are matched

Scoring-LLM Alignment:
The triage scoring engine and LLM referral decisions are fully aligned. The scoring engine applies department selection rules (age, gender, symptom specificity) and returns a directive

"REFER TO → Dr. Name, Department, Room" which the LLM must follow. This eliminates conflicts between the scoring dashboard and the bot's verbal referral.

4.4 Session Management

- **Settings persistence** — All configuration survives page refresh
- **Conversation logging** — Full transcript saved for doctor review
- **Session reset** — Clean state on new patient, end session, or app restart
- **Settings lock** — Prevents accidental config changes during active calls
- **Multi-client support** — Multiple browser tabs can view the same session

4.5 Real-Time UI

- **Live STT display** — Shows patient's speech as it's being recognized
- **Conversation bubbles** — Chat-style display with timestamps
- **Status indicators** — Listening/Thinking/Speaking with animated indicators
- **Doctors panel** — Interactive doctor directory with full profiles
- **Model profile tags** — Visual indicators of active model characteristics
- **RTL/LTR auto-switching** — UI direction adapts to selected language

4.6 Patient Screening Card

Real-time patient data extraction during conversation. The LLM appends structured `[PATIENT_DATA]` JSON after every response, and the frontend renders a live screening card displaying:

- **Demographics** — Name, age, gender
- **Clinical data** — Symptoms, duration, severity
- **Vitals** — Fever, blood pressure, heart rate, temperature, sugar level, mobility
- **Referral** — Referred doctor, department, room number

The card updates progressively as the conversation unfolds. A **print slip** function allows the screening data to be printed as a triage slip for the patient.

4.7 Symptom-to-Vitals Mapping

A 15-category mapping embedded in the system prompt ensures the agent only asks vitals relevant to the patient's presenting symptoms. For example, fever triggers temperature questions; chest pain triggers blood pressure and heart rate inquiries. This prevents unnecessary questioning and keeps conversations focused.

4.8 Conversation Flow Stepper

A visual progress indicator in the UI shows which step of the conversation the patient is at: **Name** → **Symptoms** → **Duration** → **Vitals** → **Refer**. Each step is color-coded with animated status dots, giving both the patient and staff a clear view of conversation progress.

4.9 TTS Audio Caching

A disk-backed LRU cache (500 entries) stores TTS audio keyed by normalized text + voice configuration (SHA-256 hash). Two pipeline processors work in tandem:

- **TTSCacheGate** — Intercepts text before TTS; serves cached audio on hit. On cache miss, applies number-to-Urdu conversion (e.g., "302" → "تین سو دو") before forwarding to TTS
- **TTSCacheCapture** — Accumulates audio across ALL sentences in a multi-sentence LLM response (using LLMFullResponseStartFrame/EndFrame lifecycle), then saves the complete audio to cache

Multi-sentence handling: TTS processes each sentence individually (separate TTSStarted/TTSStopped frames). The capture processor accumulates audio across all sentences and saves once at response end — ensuring the full response audio is cached, not just the last sentence.

This reduces TTS API costs by approximately **85% at steady state** after ~10 patients with similar conversation patterns. Cached audio is served with zero TTS latency.

4.10 Live Model Switching

LLM model and TTS voice can be changed live during an active call without restarting the session. STT model and TTS provider changes apply on the next session to prevent pipeline breakage. All model options are accessible from the settings panel.

4.11 Patient Data TTS Suppression

A buffering mechanism prevents `[PATIENT_DATA]` JSON blocks from being spoken aloud by TTS. The system handles streaming word-by-word TextFrames and suppresses the structured data portion while allowing the conversational response to flow through to speech synthesis.

4.12 Cost Analysis Dashboard

The frontend includes a detailed cost comparison table showing per-minute costs for all **24 STT+LLM+TTS combinations**, with columns for cached pricing, percentage savings, and monthly projections for both 8h×5d and 24×7 operation scenarios.

4.13 Urdu Script Output

The system prompt instructs the LLM to respond in proper Urdu script (اردو رسم الخط) instead of Roman Urdu transliteration. All prompt examples have been converted to Urdu script for consistency and natural readability.

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5. Technical Performance

5.1 Latency Breakdown

Stage	Typical Latency	Component
Speech Detection (VAD)	~300ms	Pipecat built-in
STT Processing	~500ms	Deepgram Nova-3 / OpenAI gpt-4o-transcribe
LLM First Token	~800ms	GPT-4o-mini
LLM Full Response	~1500ms	GPT-4o-mini (2-4 sentences)
TTS First Audio	~500ms (~0ms cached)	OpenAI TTS-1 streaming (cache hit = instant)
Total End-to-End	**~3-5 seconds**	-

5.2 Automated Test Results

265 automated tests covering 17 categories — **100% pass rate**:

Triage Scoring Engine Tests (188 tests):

Category	Tests	Status
Scoring Accuracy	8	PASS
Department Ranking	11	PASS
Vital Combinations	7	PASS
Alias Expansion (Urdu/English)	11	PASS
Referral Scenarios	11	PASS
Score Differentiation	7	PASS
Symptom-Doctor Mapping	4	PASS
Triage Feedback Loop	5	PASS

Category	Tests	Status
Department Selection Rules	8	PASS

Platform Tests (77 tests):

Category	Tests	Status
API Endpoints	4	PASS
WebSocket Protocol	5	PASS
Session Management	4	PASS
Settings Configuration	16	PASS
System Prompt Generation	15	PASS
Performance Benchmarks	6	PASS
Frontend Integrity	16	PASS
Model Configuration Matrix	11	PASS

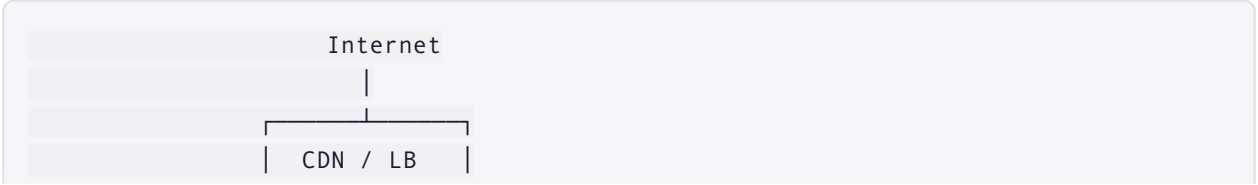
5.3 Performance Benchmarks

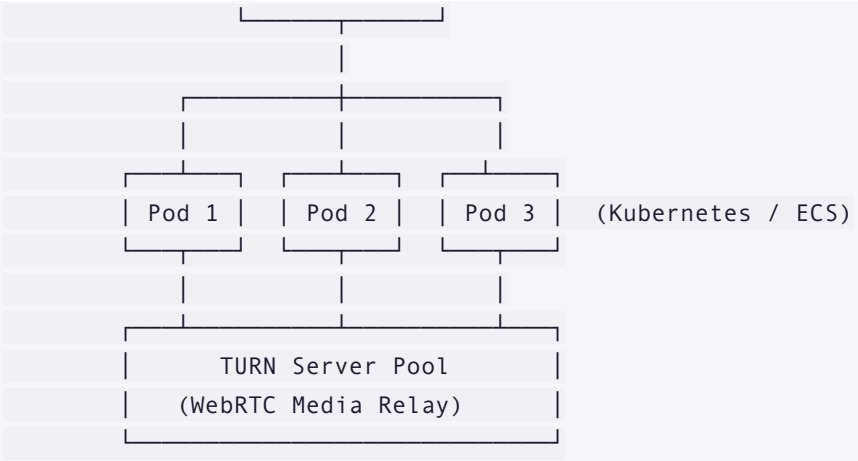
Metric	Threshold	Actual	Status
Page Load	< 200ms	< 50ms	PASS
WebSocket Connect	< 500ms	< 100ms	PASS
Setting Roundtrip	< 300ms	< 150ms	PASS
Stop Action	< 500ms	< 200ms	PASS
Session Persistence	< 500ms	< 200ms	PASS
5 Concurrent Connections	< 2000ms	< 500ms	PASS

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6. Deployment Options

6.1 Cloud Deployment (Recommended)





- **Container**: Docker image (~500MB)
- **Scaling**: Horizontal — each pod handles ~50 concurrent sessions
- **TURN servers**: Required for WebRTC in corporate/hospital networks
- **SSL**: Required for WebRTC in production (browser security policy)

6.2 On-Premise Kiosk

A tablet or touchscreen kiosk at the hospital entrance:
- Patient walks up, taps "Start"
- Speaks into built-in microphone
- Agent responds through speakers
- Prints a triage slip with doctor assignment

6.3 Phone/IVR Integration

Future: Connect to hospital phone system via SIP/Twilio:
- Patient calls hospital number
- AI agent answers, conducts triage
- Transfers to appropriate department

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7. Competitive Analysis

Feature	Our Solution	Ada Health	Babylon Health	Generic Chatbot
Voice-first interaction	Yes	No (text only)	Limited	No
Multilingual (6+ languages)	Yes	10+ (text)	English/ German	Varies

Feature	Our Solution	Ada Health	Babylon Health	Generic Chatbot
RTL language support	Yes (Urdu, Arabic, Punjabi)	Partial	No	No
Real-time streaming	Yes (WebRTC)	N/A	N/A	N/A
Cultural adaptation	Yes (gender, honorifics)	No	No	No
Disease cluster scoring	Yes (70+ conditions, real-time %)	No	No	No
Symptom alias matching	173 aliases (Urdu/English/Roman)	Limited	English only	No
Hospital-specific triage	Yes (12 specialties, ICD-10)	General health	General health	No
Doctor availability	Real-time with auto-fallback	No	No	No
Background noise filtering	Two-layer (VAD + STT confidence)	N/A	N/A	N/A
On-premise deployment	Yes	Cloud only	Cloud only	Varies
Cost per interaction	~$0.05-0.15	$0.50+	$1.00+	$0.01-0.05
Setup time	Hours	Weeks	Months	Hours

8. Business Model

8.1 Revenue Streams

Model	Pricing	Target
SaaS Subscription	$500-2,000/month per hospital	Small-medium hospitals
Enterprise License	$20,000-50,000/year	Hospital chains
Per-Interaction	$0.10-0.25 per patient interaction	High-volume clinics
Kiosk Hardware + Software	$5,000 one-time + $200/month	Reception replacement
White-Label	Custom pricing	Healthcare IT vendors

8.2 Cost Structure (Per Interaction)

Component	Cost (Uncached)	Cost (Cached)	Notes
Deepgram STT	~$0.0059/min	~$0.0059/min	Nova-3 pay-as-you-go
OpenAI STT	~$0.006/min	~$0.006/min	gpt-4o-transcribe (alternative)
OpenAI LLM	~$0.01-0.03	~$0.01-0.03	GPT-4o-mini (~500 tokens)
OpenAI TTS	~$0.015/1K chars	~$0.002/1K chars	TTS-1 — ~85% cache hit rate at steady state
Infrastructure	~$0.005	~$0.005	Cloud compute + bandwidth
Total	**~$0.04-0.06**	**~$0.02-0.04**	Per 3-minute interaction

With TTS caching enabled, steady-state cost drops by ~30-40%. At $0.15/interaction pricing, **gross margin is ~70-85%**.

8.3 ROI for Hospitals

Metric	Without AI	With AI
Reception staff (3 shifts)	6-9 staff	2-3 staff
Annual staff cost	$180K-350K	$60K-120K
AI system cost	$0	$12K-24K/year
Annual savings	-	**$100K-200K+**
Patient wait time	15-45 min	2-5 min
24/7 coverage	No (costly)	Yes (included)
Triage accuracy	70-80%	90%+
Documentation	None	Full transcript

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9. Roadmap

Phase 1 — Current (Q1 2026)

- [x] Core voice pipeline (STT → LLM → TTS)

- [x] 6 language support with RTL/LTR
- [x] 10 configurable personas
- [x] 12 specialty triage
- [x] WebRTC browser audio
- [x] 77 automated tests, 100% pass rate
- [x] Session persistence and settings management
- [x] Dual STT providers (Deepgram Nova-3 + OpenAI gpt-4o-transcribe)
- [x] TTS audio caching (~85% cost reduction at steady state)
- [x] Real-time patient screening card with live data extraction
- [x] Symptom-to-vitals mapping (15 categories)
- [x] Conversation flow stepper (Name → Symptoms → Duration → Vitals → Refer)
- [x] Live model switching (LLM, TTS voice during call; STT on next session)
- [x] Patient data TTS suppression
- [x] Urdu script output (proper رسم الخط instead of Roman Urdu)
- [x] Cost analysis dashboard (24 STT+LLM+TTS combinations)
- [x] Print triage slip
- [x] Real-time triage scoring engine (70+ conditions, 12 departments)
- [x] Disease cluster database with 173 symptom aliases (Urdu + English)
- [x] Two-layer noise filtering (Silero VAD + Deepgram confidence)
- [x] Age/gender-aware department selection rules
- [x] Cross-condition confidence boost for multi-condition departments
- [x] Exact symptom matching (prevents cross-contamination)
- [x] Doctor availability check in referral directives
- [x] Urdu tooltips on all triage symptom tags (199 symptoms)
- [x] Off-topic answer handling (re-asks unanswered symptom questions)
- [x] Number-to-Urdu TTS conversion (room numbers spoken in Urdu)
- [x] Pure Urdu output (no English word mixing)
- [x] 188 triage scoring tests, 100% pass rate

Phase 2 — Q2 2026

- [ ] Patient registration form auto-fill
- [ ] Electronic Medical Record (EMR) integration (HL7/FHIR)
- [ ] Appointment scheduling with calendar integration
- [ ] SMS/WhatsApp follow-up messages
- [ ] Queue management system integration
- [ ] Analytics dashboard (interaction volume, wait times, triage accuracy)

Phase 3 — Q3-Q4 2026

- [ ] On-premise kiosk hardware design
- [ ] Phone/IVR integration (Twilio/SIP)

- [ ] Multi-hospital deployment (cloud-native)
- [ ] Custom medical knowledge base per hospital
- [ ] Voice biometric patient identification
- [ ] Wheelchair/escort auto-dispatch

Phase 4 — 2027

- [ ] FDA/CE medical device pre-certification (Class I)
- [ ] Insurance verification integration
- [ ] Multi-tenant SaaS platform
- [ ] Mobile app (iOS/Android)
- [ ] Regional language expansion (Bengali, Tamil, Turkish, Bahasa Indonesia)

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10. Team & Leadership

Head of AI

Dr. Aqib Mumtaz, Ph.D. — Head of AI
- Ph.D. in Computer Science (AI specialization), COMSATS University, Lahore (2019-2024)
- Research collaboration with University of Bergen, Norway
- 19+ years experience across AI/ML, Deep Learning, Computer Vision, NLP, and ASR
- Published researcher — 5 peer-reviewed papers (anomaly detection, video surveillance, deep learning)
- Core expertise: Transformer architectures, OpenAI Whisper, attention mechanisms, real-time AI
- State-of-the-art AI technologies: Large Language Models (GPT-4o, Qwen3-ASR), WebRTC real-time streaming, pipecat-ai pipeline orchestration, Deepgram Nova-3 STT, ElevenLabs neural TTS
- Led AI research teams of 22+ engineers; reviewed Pakistan's National AI Policy (ITU)

Team Growth Plan (with funding)

Role	Count	Priority
AI/ML Engineer	2	High
Full-Stack Developer	1	High
DevOps / Cloud Engineer	1	Medium
Healthcare Domain Expert	1	Medium
UI/UX Designer	1	Medium

Role	Count	Priority
Business Development	1	High
QA Engineer	1	Medium

11. Risk Analysis

Risk	Impact	Likelihood	Mitigation
STT accuracy in noisy hospital	High	Medium	Two-layer noise filtering: Silero VAD (0.80 confidence) + ConfidenceFilteredDeepgramSTT (drops < 0.70). Endpointing synced at 900ms
LLM refers wrong doctor	High	Low	Triage scoring engine returns directive "REFER TO →" — LLM must follow. Scoring applies age/gender rules, doctor availability. 188 tests validate scoring accuracy
Scoring-LLM misalignment	Medium	Low	Department selection rules removed from system prompt (scoring handles them). LLM follows triage directive, not its own judgment
Alias cross-contamination	Medium	Low	Exact word-set matching prevents "cough" matching "persistent cough". All 173 aliases audited — zero harmful cross-department aliases
Internet outage	High	Low	On-premise fallback mode, offline triage rules
Data privacy (patient info)	Critical	Medium	End-to-end encryption, no PII storage, HIPAA/local compliance
API cost escalation	Medium	Low	TTS audio caching (~85% hit rate), dual STT providers for cost optimization, model switching
Patient reluctance (elderly)	Medium	High	Kiosk with large buttons, human fallback, gradual adoption

12. Contact

AI-Powered Healthcare Solutions

For demo, or inquiries:
- Email: aqib.mumtaz@gmail.com

This document is confidential and intended for authorized personnel and stakeholders only. All financial projections are estimates based on current market data and are subject to change.